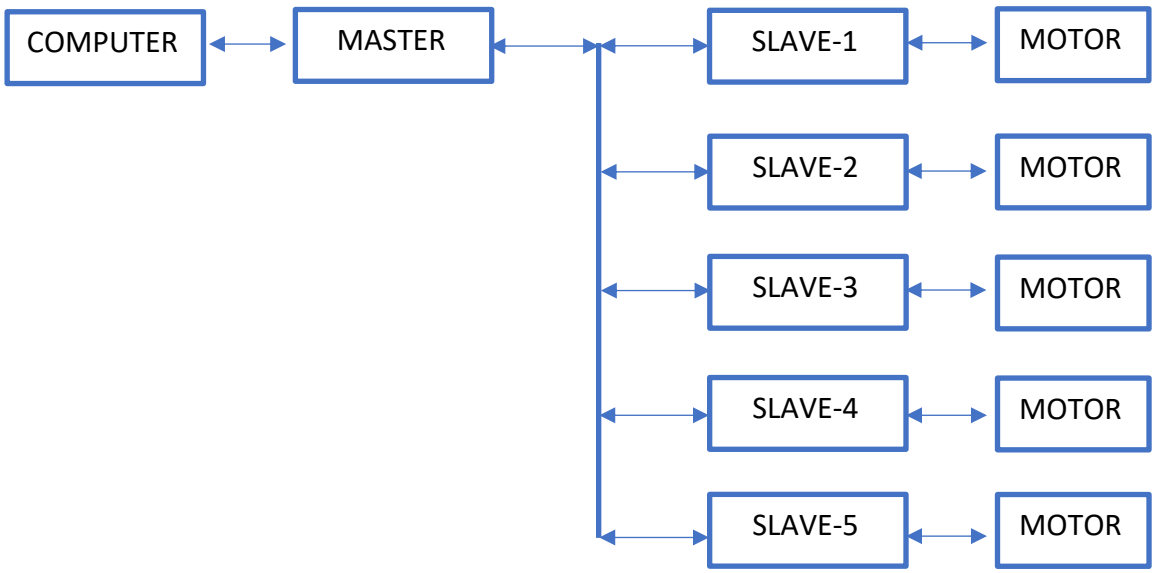
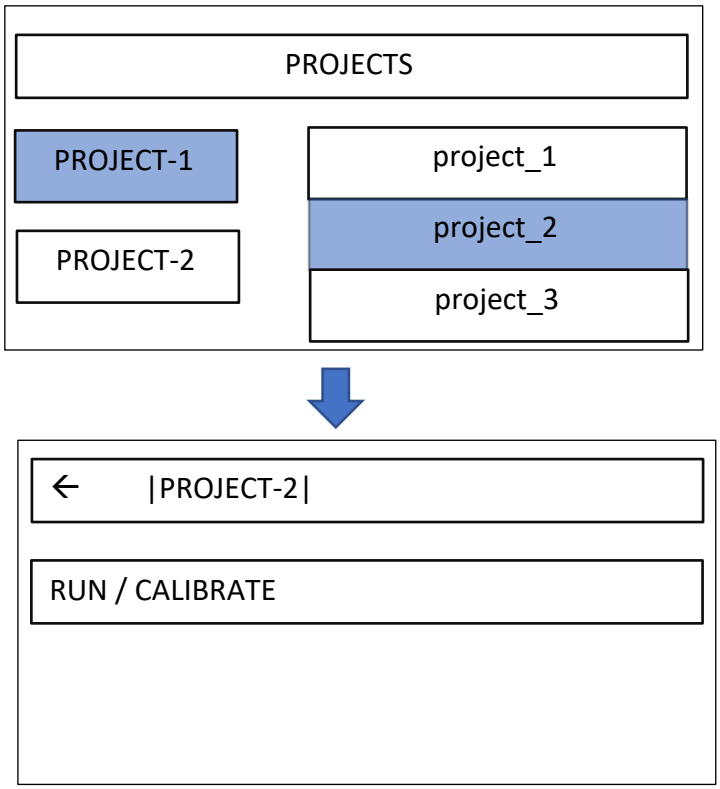


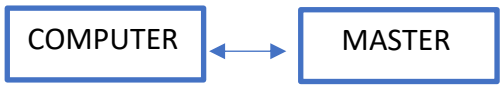
COMPUTER

1- DESKTOP/FOLDER: KURVATUR
    FOLDER-1: CALIBRATION
        FILE_NAME: calibration_values.txt
    FOLDER-2: PROJECTS
        FOLDER: PROJECT_1
            FILE_NAME: project_1_position_values.txt
            FILE_NAME: project_2_position_values.txt
            FILE_NAME: project_3_position_values.txt

        FOLDER: PROJECT_2
            FILE_NAME: project_1_position_values.txt
            FILE_NAME: project_2_position_values.txt
2- COMPUTER/FOLDER: TEST_HISTORY
    FOLDER: DATE/MOUTH/YEAR
        FILE: PROJECT_NAME

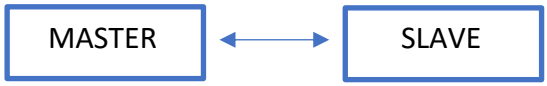
3- SOFTWARE





PROCESS:

1- If motor last position not equal calibration_values.txt, make button name “CALIBRATE”
2- If motor last position equal calibration_values.txt, make button name “RUN”
3- If temperature out of the limitation, error display “temperature = ”
4- If humility out of the limitation, error display “humility =”
5- If motor id not found, error display “motor id not found = ”



MASTER

CHECK (CONSISTENTLY):

(if out of the limitations: stop all slave movement, send the error to software)

1-Motor Position
2-Temperature
3-Humidity
4-Motor ID (coming from slave PCB)

SAVE (CONSISTENTLY):

(FOLDER: computer/TEST_HISTORY/DATE_MONTH_YEAR/PROJECT_NAME.TXT)
(Master Board)

1-The last position value of motors
2-The last activated time: second/minute/hour/day/month/year
3-The last temperature value
4-The last humility value

PROCESS:

1- If motors are in the project position:
   1.1-Get the values: desktop/kurvatur/calibration/calibration_values.txt
   1.2-Transfer the values position_values_array (ARRAY)
   1.3-Send it the slave PCB
2- If motors are in the calibration position:
   2.1- Get the project name: PROJECT_1/ project_2_position_values.txt
   2.2- Transfer the values position_values_array (ARRAY) from project_2_position_values.txt
   2.3- Send it to slave PCB

3- Save all movements computer/TEST_HISTORY/DATE_MONTH_YEAR/PROJECT_NAME.TXT
4- Save last values in the master board

SLAVE

DEFINE

(dynamic variable, it has to be changeable):

1-MOTOR ID

CHECK (CONSISTENTLY):

(if out of the limitations: stop all slave movement, send error to master board)

1-Motor Position
2-Temperature
3-Humidity

SAVE (CONSISTENTLY):

(Slave Board)

1-The last position value of motors
2-The last activated: second/minute/hour/day/month/year

3-The last temperature value
4-The last humidity value

PROCESS:

1- get the value from ARRAY (coming from MASTER BOARD) (position_values_array)
2- get the position value based on PCB ID
3- drive the stepper motor to the position
4- send the last position value to the master board

5- save the last values in slave board